

Assignment-3 Session 3

Question 1: Download any free APK file (for example, a Calculator app) and install it on your Android emulator using Android Studio Device Manager.

Answer

Application Installed

Application Name: Calculator APK

Steps Performed

1. Downloaded the APK file.
2. Opened Android Studio.
3. Started the Android Emulator using Device Manager.
4. Dragged and dropped the APK file onto the emulator (or installed it using the Install APK option).
5. Waited for the installation to complete.
6. Verified that the application icon appeared in the app drawer.
7. Opened the application successfully.

Result

The APK installed successfully and launched without installation errors.

Question 2: Rotate the device from Portrait to Landscape and back. Note any UI issues.

Answer

Orientation	Observation
Portrait	UI displayed correctly with all buttons visible.
Landscape	Buttons appeared closer together, but remained usable.
Back to Portrait	Layout returned correctly without overlapping elements.

Sample UI Observations

- 1) Button spacing became smaller in landscape mode.
- 2) Text alignment remained correct after rotation.
- 3) No application crash occurred during orientation changes.

Result

The application handled screen rotation successfully with only minor layout adjustments.

Question 3: Turn off the internet connection in the emulator and test a feature that requires a network.

Answer

Feature Tested: Login

Steps

1. Disabled network connectivity in the emulator.
2. Opened the application.
3. Attempted to log in using valid credentials.

Observation

The application displayed the following message:

"No Internet Connection. Please check your network and try again."

Result

The application handled the offline scenario gracefully by showing an appropriate error message instead of crashing.

Question 4: Test the Login or Registration flow on both the emulator and a real Android device. Compare the experience.

Answer

Note: Replace these observations with your own after testing.

Feature	Emulator	Real Device
Login Speed	Slightly slower due to virtualization.	Faster and more responsive.
UI Rendering	Correct but slightly less smooth.	Smoother animations and transitions.
Notifications	May be delayed or unavailable depending on emulator configuration.	Delivered normally and promptly.

Differences Observed

- The real device responded more quickly than the emulator.
- UI animations were smoother on the real device.

Question 5: Describe how you would test push notifications using both an emulator and a real device. Mention at least one limitation of emulators.

Answer

Application: Instagram

Testing on Emulator

- 1) Install the application.
- 2) Sign in with a test account.
- 3) Enable notification permissions.
- 4) Trigger a notification (for example, send a message or like a post from another account).
- 5) Verify whether the notification is received and opens the correct screen.

Testing on a Real Device

1. Install the application.
2. Enable notification permissions.
3. Trigger different notification types (messages, likes, comments, follow requests).
4. Verify notification content, sound, vibration, and navigation.
5. Confirm notifications are received in both foreground and background states.

Limitation of Emulators

Emulators may not reliably reproduce real push notification behavior because they can differ from physical devices in services such as Google Play Services, background processing, battery optimization, and manufacturer-specific notification handling. Therefore, notification testing should always be validated on at least one real Android device.